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Tailored bamboo fractionation with ternary deep eutectic solvent system to maximize biorefinery toward enzymatic saccharification and lignin recovery

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ABSTRACT

Six different ternary deep eutectic solvent (DES) of tetraethylammonium chloride (TEAC)/oxalic acid (OA)-polyols were synthesized to improve the production of fermentable sugars and lignin by-products. The bamboo fractionation and bioconversion of cellulose residues can be enhanced by the ternary DES pretreatment, and its increasing efficiency was positively related to the alkyl carbon chain length and hydroxyl number of polyols in the hydrogen bond donor. The 1,4-butanediol (BDO)-assisted DES pretreatment showed excellent performance in achieving high xylan removal (86.7 %) and delignification (91.98 %). The pretreated bamboo was used for enzymatic hydrolysis and ethanol fermentation, with glucose yield and ethanol production of 76.13 % and 16.17 g/L, which was significantly improved by 30 % and 48 % compared to bamboo treated with corresponding binary TEAC/OA. The characterization of trinary pretreated bamboo revealed that the addition of polyols helped to protect the β-O-4 substructure of lignin, reduce the recondensation of lignin fragments, and thus enhance enzyme digestibility. Furthermore, 37.9% of lignin with high purity and highly active hydroxyl substructures was recovered. The proposed novel TEAC/OA-BDO DES mixture has great potential as an effective pretreatment solvent for future bamboo biorefinery.

1. Introduction

With the reduction of resources and global warming, it is necessary to find and develop new renewable energy sources with minimal carbon emissions. Biomass biorefinery is of key significance to support the sustainable and recyclable bioeconomy. Bamboo has the characteristics of fast growth, high mechanical strength, and low management requirements. There are approximately 5 million hectares of bamboo forests in China, which is about 3 % of the total forest area [1]. During the bamboo processing, about 30–40 % of the processing residue in the entire bamboo was generated. These residues consisted of cellulose, hemicellulose and lignin and can be used to produce chemicals, biofuels, and functional materials, which will be beneficial for the sustainable development of bamboo biorefinery in southwestern China.

Pretreatment technology, as a key and initial step during the biomass biorefinery, have been developed to overcome the natural recalcitrance and highly lignified structure properties of bamboo [2,3]. Among existing pretreatment techniques, deep eutectic solvent (DES) has potential applications in selectively removing lignin, and retaining cellulose and hemicellulose. DESs as a green solvent has the advantages of simple preparation, low cost, nontoxicity, and environmental friendliness, and can be classified into alkaline, acidic, and neutral system according to the specific characteristics of hydrogen bond donors (HBD) [4,5]. Acid-based DESs using organic carboxylic acid as HBD have been proven to be more effective than alkaline and neutral DES in biomass fractionation [6]. The carboxyl groups in the acidic DES provided active hydrogen protons to catalyze the cleavage of ether bonds and LCC structure in lignocellulose [7,8]. However, under strong acid or high temperature conditions, lignin retained on cellulose substrates and extracted lignin, typically undergoes severe and irreversible

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repolymerization [9], similar to conventional acid pretreatment. This severe condensation reaction will affect the subsequent conversion of carbohydrates into fermentable sugars and is also detrimental to further valorization and applications of regenerated lignin, which will also limit the integration of acidic DES systems into current biorefining processes of biomass.

In recent years, in order to modify and optimize the performance of acid-based DES, the synthesis of multicomponent DES systems and their advantage over the binary counterparts have received extensive attention. Liu et al. [10] added ethylene glycol (EG) to ChCl/OA to obtain lignin with high β-O-4 content and high-quality cellulose. When different diols were added to the system of ChCl and OA for the fractionation of lignocellulosic biomass, a separation efficiency of 85 % was achieved. They found that polyols can be as solvent to protect the structure of lignin [11]. Glycol-assisted DES pretreatment can improve the removal of lignin and xylan, and enzymatic saccharification during bamboo pretreatment [12]. The -OH group in polyols played a vital role in the hydrogen bonding strength of the whole DES system, which has an important impact on biomass pretreatment. However, there is a lack of systematic analysis on how polyols assist acidic DES in inhibiting lignin condensation, as well as the effects of polyols with different alkyl chains and hydroxyl contents on the performance of acidic DES pretreatment. In addition, compared with the commonly used hydrogen bond acceptor (HBA) choline chloride, tetraethylammonium chloride (TEAC) is an inexpensive and readily available HBA that can also be used during the synthesis of DES solvents for lignocellulose pretreatment.

In this work, six different polyols were used to synthesize ternary DES systems for further bamboo fractionation. The DES pretreatment performance was evaluated by measuring the changes in chemical composition and enzymatic digestibility after treatment. And the synergistic mechanism of ChCl/OA and polyol is also elucidated through the substrate's characteristic and molecular dynamics simulation. Furthermore, the structural characteristics of regenerated-lignin were thoroughly characterized to investigate their potential for further valorization. The designed DES mixture will provide a sustainable solvent platform for lignocellulose biorefineries and achieve higher biorefining efficiency.

2. Materials and methods

2.1. Feedstock and chemicals

The raw material *Dendrocalamus giganteus* (*D. giganteus*, DG), Dendrocalamus sinicus (*D. sinicus*, DS), Neosinocalamus affinis (*N. affinis*, NA), and Dendrocalamus hamiltonii (D. hamiltonii, DH) were obtained from a bamboo processing factory in Kunming (Yunnan, China); the bamboo residues were grinded and sieved to 40–60 mesh. Tetramethylammonium chloride (TEAC, ≥98 %), oxalic acid (OA, ≥99.5 %), ethanol (EtOH), ethylene glycol (EG), propanol (PrOH), glycerol (GLY), butanol (BuOH) and 1,4-butanediol (BDO) were bought from Yunnan Sinopharm Chemical Reagent Co., Ltd. Cellulase (Cellic CTec2, 200 FPU/mL) and Celluclast 1.5 L (40 mg/mL) were obtained from Sigma-Aldrich (Shanghai, China). *Saccharomyces cerevisiae* used for ethanol fermentation were achieved from Yichang Angel Yeast Company.

2.2. DES pretreatment of bamboo residues

Acidic DES was prepared by mixing TEAC and OA in a molar ratio of 1:1. Then EtOH, EG, PrOH, GLY, BuOH and BDO were used as the second HBD, respectively. Mix them with the binary TEAC/OA at a volume fraction of 20 % to prepare the polyol-assisted ternary DES. The binary and ternary mixtures were magnetically stirred for 30 min at 60 °C until clear and uniform liquids formed. The Kamlet-Taft parameters (α, β, and π*) of these DESs were measured with 4-nitroaniline, N, N-diethyl-4-nitroanili, and Nile red as indicators [13].

Bamboo powder was added into the binary and ternary DES systems at a solid-liquid ratio of 1:10 (*w/v*), then stirred at 120 °C for 7 h. Filter the solid residues, wash by hot water until the pH value was neutral, then dry for further component analysis. The optimization of TEAC/OA-BDO pretreatment conditions is shown in the supplementary materials. The filtrate was added to 300 mL ethanol/water (3:1) to obtain regenerated-lignin. Chemical component of bamboo residue was determined according to the NREL standard method [14]. The following equations were used to calculate the solid yield, glucan recovery, and delignification after pretreatment:

$$\text{Solid yield}\ (\%) = \frac{\text{treated sample weight}}{\text{raw material weight}} \times 100 \quad (1)$$

$$\text{Glucan recovery}\ (\%) = \frac{\text{glucan weight in treated sample}}{\text{glucan weight in raw material}} \times 100 \quad (2)$$

$$\text{Delignification}\ (\%) = \left(1 - \frac{\text{lignin weight in treated sample}}{\text{lignin weight in raw material}}\right) \times 100 \quad (3)$$

2.3. Enzymatic digestibility and fermentation of pretreated bamboo residues

Untreated bamboo and pretreated substrates were subjected to enzymatic hydrolysis with Cellic CTec2 enzyme (20 FPU/g substrate) at 5 % solid concentration (*w/v*) in sodium citrate buffer (pH 4.8). All substrates were hydrolyzed in a shaker at 50 °C and 150 rpm for 72 h. The hydrolysis reaction solution at different times was sampled to determine the amount of released glucose and xylose. After sterilizing the obtained hydrolysate at 121 °C for 20 min, incubate it with *Saccharomyces cerevisiae* at 35 °C and 90 rpm for 24 h. Regularly collect the supernatant to analyze the ethanol content. The analysis of sugars and ethanol were determined using high performance liquid chromatography (HPLC, Agilent 1260, USA) with a Bio-Rad Aminex HPX-87H column. The enzymatic saccharification yield and ethanol production were calculated based on our previous publications [15,16].

2.4. Cellulase adsorption of treated D. giganteus

The adsorption of Celluclast 1.5 L (45 mg/mL of protein content) on the pretreated samples was analyzed using the Langmuir adsorption isotherm. Various cellulase concentrations (0.01–1.0 mg/mL) were mixed with pretreated bamboo and incubated in 5 mM sodium citrate buffer for 3 h. After reach equilibrium, the protein contents in the supernatant were measured by the Bradford method. Nonlinear regression was performed on adsorption data by the Langmuir isotherm ($\Gamma = \Gamma_m KC/(1 + KC)$), where C represented the free cellulase content in solution (mg/mL). And adsorption parameters including the Langmuir constant (K), maximum adsorption capacity (Γ_m) and distribution coefficient ($R = \Gamma_m \times K$, L/g) were determined [15,16].

2.5. Characterization of treated D. giganteus

The hydrophobicity and enzyme accessibility of samples were measured by Rose Bengal dye and direct red 28 staining method described [17,18]. The degree of polymerization (DP) was obtained through measuring the intrinsic viscosity of fiber solution in copper ethylenediamine solvent (CED). The BET surface area of pretreated bamboo was analyzed with the ASAP2460 specific surface area instrument (Micromeritics Instrument Ltd., USA). The crystallinity index (CrI) of raw bamboo and pretreated bamboo was determined with XRD, and the actual effect of DES pretreatment on CrI was expressed as the ratio of CrI/cellulose content [19]. XPS analyzer (Thermo Scientific, USA) was used to measure the distribution of C and O elements on the pretreated substrates. The surface coverage of lignin and carbohydrate was calculated based on our previous studies. The substrate morphology was

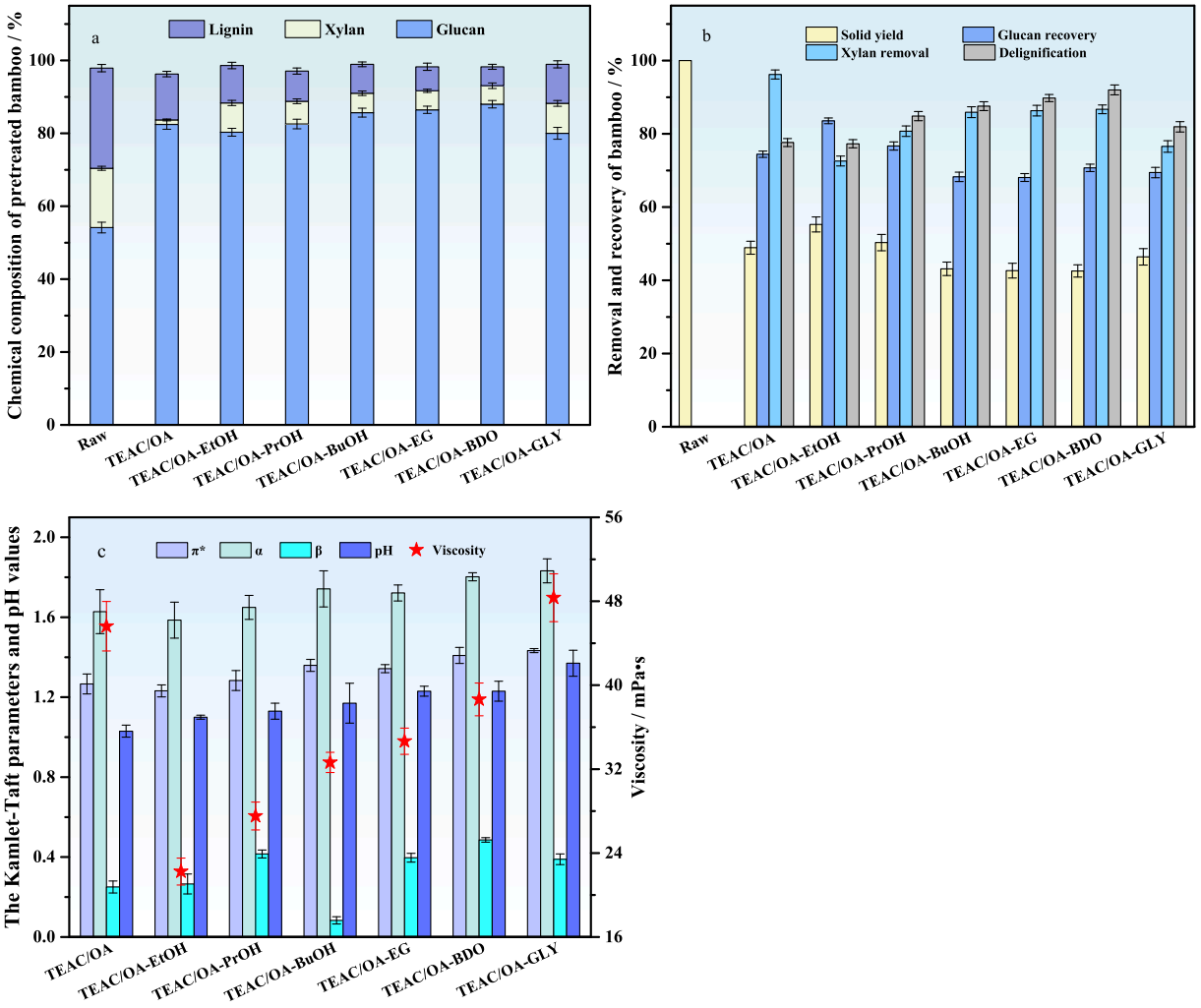


Fig. 1. Component changes of bamboo residues under various DES systems (a and b); and the Kamlet-Taft empirical parameters of different DES systems.

observed by scanning electron microscopy (SEM). FTIR spectrophotometer recorded in the range of 4000–400 cm^{-1} was used to describe the structure characteristics of bamboo. The 2D HSQC and ^{31}P NMR of regenerated lignin were recorded by 500 MHz NMR spectrometer (Bruker, Germany); the sample preparation and operation methods were based on previous studies [20].

2.6. Computational analysis

The structure of all molecular were optimized at the B3LYP-D3 (BJ)/def2-SVP level using ORCA 4.0 software [21]. The electrostatic potential distribution (ESP) and reduced density gradient (RDG) of all molecules were analyzed with the Multiwfn program, and the VMD program was used for visualization [22]. Classic molecular dynamics simulations were performed to study the mixed solution from the atomic level [23,24]. Veratrylglycerol-β-guaiacol ether (VG), 4-*O*-methyl Gluconic acid xylose and cellobiose were chosen as the lignin, hemicellulose, cellulose models, respectively.

3. Results and discussion

3.1. The fractionation of bamboo residues in the polyol-assisted ternary DES

Six different ternary DESs were synthesized to study the influence of polyol on the DES pretreatment of bamboo residues. As shown in Fig. 1a, the raw material *D. giganteus* (DG) used in this study contained 54.16 % glucan, 16.25 % xylan, and 27.48 % lignin. When DG was treated with the binary DES of TEAC/OA at 120 °C for 7 h, and the solid yields, the removal of xylan and lignin were 48.9 %, 96.21 % and 77.63 %, respectively. This suggests that the binary DES is more effective in removing the lignin and xylan of bamboo residues. Because the active proton from acid-based DES system can catalyze the breaking of lignin ether bonds and the lignin-carbohydrate complexes, further leading to the dissociation of bamboo recalcitrant structures [25]. Similar to choline chloride, TEAC is also an inexpensive and easily available HBA that has been reported and used for synthesizing DES solvents. The TEAC structure has a shorter alkyl chain as compared to choline chloride. It was reported that DES composed of HBA with short alkyl chains is more effective in removing lignin and xylan during the pretreatment [26,27]. Considering that during the acidic DESs pretreatment, pseudo-lignin can form on the surface of the substrate, and regenerated lignin may also undergo severe condensation, TEAC/OA will be mixed with various polyols in a certain molar ratio to form a new DES system for alleviating these negative impacts. The addition of EtOH and PrOH to the binary DES has no significant effect on the removal of lignin and xylan. However, it can be clearly observed from Fig. 1 that with the extension of carbon chain and hydroxyl content in the third component structure, the ternary DES pretreatment gave significantly lower solid yields of 43.14 % for TEAC/OA-BuOH, 42.64 % for TEAC/OA-EG, 42.55 % for TEAC/OA-BDO, and 46.41 % for TEAC/OA-GLY, respectively. For these four DES systems, the delignification significantly improved, up to 87.55 %,

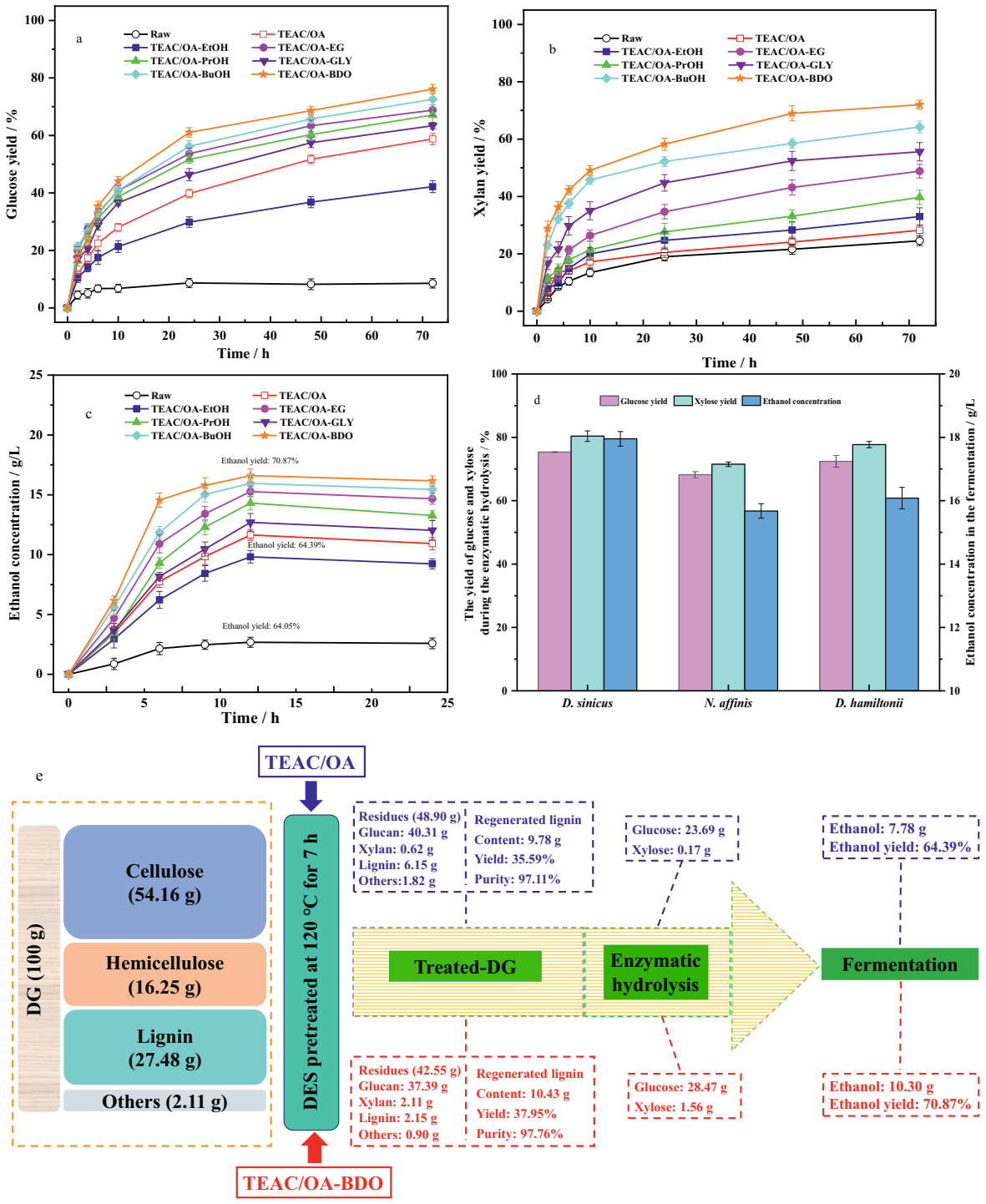


Fig. 2. Enzymatic digestibility of (a) glucan, (b) xylan and ethanol production (c) of the pretreated bamboo; (d) comparison of enzymatic hydrolysis and fermentation performance of three bamboo; (f) mass balance of the TEAC/OA and TEAC/OA-BDO DES pretreatment process, enzymatic hydrolysis and microbial fermentation.

89.77 %, 91.98 % and 81.91 % respectively, while the yield of hemicellulose removal decreased to about 80 %. This may be due to the addition of the third component polyol affecting the strength of the hydrogen bonds and acidity of the DES mixture. The obvious decrease of xylan removal in the ternary DES is because the addition of the third component diluted the H^+ concentration in TEAC/OA system. Moreover, the lignin removal gradually increased with the extension of the alkyl carbon chain length in the tertiary component. Fig. 1b showed that the delignification of DG in the TEAC/OA-PrOH (84.84 %) and TEBAC/OA-BDO (91.98 %) were higher than that of the TEAC/OA-EtOH (77.28 %). And the influence of hydroxyl number on the pretreatment performance of TEAC-based tertiary DES with the same length of alkyl chain was analyzed. Lignin removal yield in TEAC/OA-EtOH was lower than that in TEAC/OA-EG. Similarly, the delignification of TEAC/OA-BuOH was lower than that of TEAC/OA-BDO. The excess -OH of alkyl alcohols as HBDs in DES formed more hydrogen bonds with binary TEAC/OA, competing with the original hydrogen bond network in bamboo, ultimately leading to more severe hydrogen bond destruction in DG.

Wang et al. [28] found that the presence of hydroxyl groups in the third component of DES enhanced the solubility of bamboo lignin. The removal of xylan and lignin by TEAC/OA-GLY decreased to 76.55 % and 81.91 %, respectively, which may be related to the high viscosity of GLY-based DES. In addition, the cellulose retention rate was >80 % after DES pretreatment, and the high cellulose recovery was beneficial for subsequent enzymatic hydrolysis.

Among them, 1,4-butanediol (BDO) plays a relatively significant role in the bamboo fractionation than other polyol components. To better understand the impact of BDO on the TEAC/OA pretreatment of bamboo, the Kamlet-Taft empirical parameters were determined, including hydrogen bond acidity α, basicity β and dipolarity/polarizability π*. The α and β values of TEAC/OA were only 1.628 and 0.25, respectively (Fig. 1c). The α values of TEAC/OA-EtOH, TEAC/OA-PrOH, and TEAC/OA-BuOH were 1.586, 1.649 and 1.742, respectively. And as the hydroxyl group numbers increased, the α value also increased. TEAC/OA-BDO reached a maximum value of 1.78, indicating that it easily released H^+. The β value also increased with the length of carbon chain and the number of hydroxyl groups in the third component structure, and the TEAC/OA-BDO achieved up to 0.486. The π* values also showed a slight increase. This indicated that the addition of BDO to binary DES can improve the H-bond acceptance ability and acidic strength of the solvent, making it easier to form strong H-bonds with LCC, thereby improving the separation of biomass components. Satisfactorily, the addition of a third component decreased the viscosity of binary TEAC/OA DES from 45.62 mPa·s to 38.65 mPa·s. In addition, the viscosity of TEAC/OA-GLY was 48.34 mPa·s, slightly higher than that of the binary TEAC/OA system, which inevitably weakened the mobility of the molecular compounds.

3.2. Enzymatic saccharification and fermentation of different bamboos pretreated with ternary DES

The enzymatic digestibility was used to examine the effect of DES pretreatment on the bioconversion of bamboo residues, the yields of sugars and ethanol were depicted in Fig. 2. The residues after TEAC/OA pretreatment exhibited a glucose yield of 58.77 % after 72 h of enzymatic saccharification, which was about 7 times higher than raw DG (8.57 %). The significant improvement of enzymatic saccharification was attributed to the destruction of lignocellulosic structure during the DES pretreatment, with most lignin and hemicellulose being removed. However, although a certain amount of lignin and xylan was removed during the TEAC/OA pretreatment, the hydrolysis yield of TEAC/OA-DG (58.77 %) was relatively low compared to other pretreated substrates with the same lignin and xylan content. This meant that there may be barrier components on the surface of pretreated samples, which can affect the accessibility of cellulase to the substrates. It was reported that the dissolved lignin may redistributed or undergo further condensation on the surface of the loosed fiber during the acidic DES pretreatment, which can reduce the binding sites between cellulose and cellulase. And the content and properties of the surface lignin played a more important role than the total lignin content during the enzymatic saccharification [29,30]. In comparison, the enzymatic saccharification of solid residues from TEAC/OA-PrOH and TEAC/OA-BuOH pretreatment were enhanced to 67.09 % and 72.58 %, which was about 20 % increase compared with that of sample treated TEAC/OA. When ethanol and binary TEAC/OA formed a ternary DES system, the glucose yield of the substrates decreases to 42.20 %. Furthermore, it can be seen that there was a certain relation between the glucose yield and the carbon chain length of polyol in the ternary DES system. The longer the carbon chain of the polyol, the higher the glucose yield. The low glucose yield of TEAC/OA-EtOH was possibly because of the reduction in the acidity of DES system. Xylose yields of 28–64.22 % for pretreated bamboo with binary and ternary DESs were also obtained after saccharification, albeit with less retention of xylan. In addition, as the hydroxyl content of the third component in DES solvent system increased, the enzymatic hydrolysis efficiency was enhanced. The 72 h glucose yield of bamboo pretreated with TEAC/OA-EG (68.81 %) increased by 63 % compared to that of TEAC/OA-EtOH. TEAC/OA-BDO showed the highest enzymatic hydrolysis yield of 76.13 %, which was 30 % higher than that of binary TEAC/OA samples. These results indicated the importance of HBD and HBA structures in the biomass pretreatment with DES. It was observed that the glucose yield of TEAC/OA-GLY pretreated DG (63.38 %) went against the increasing trend, which may be due to its high viscosity. The high viscosity of DES made it adhere to the treated bamboo surface. Interestingly, there were no visible differences in the removal of lignin and xylan between TEAC/OA and TEAC/OA–BDO pretreatment, but the glucose yield of the substrates treated with ternary DES was about 30 % higher than that of the binary treated DG. This may be related to the influence of the pseudo-lignin and condensed lignin of the substrate surface on the enzymatic activity and cellulose accessibility of binary pretreated samples. Free hydroxyl groups in the polyols can interact with the free and etherified hydroxyl groups of lignin, resulting in high lignin fractionation yield and improved the cellulose accessibility to cellulase [31,32]. Previous studies have shown that surface lignin on pretreated substrates can be removed by washing the solid with organic solvents and water [15,33]. After washing the pretreated sample with sodium carbonate, lignin was further decreased by 21 % compared to the water wash [33]. Although the combination of acidic DES pretreatment and organic solvent washing can lead to a reduction in the content of lignin deposition and surface lignin, polyols can directly alleviate this inhibitory effect during DES pretreatment, which is beneficial for the industrialization of DES pretreatment.

Continuously, the hydrolyzed supernatants were fermented with yeast to produce ethanol. The glucoses were almost consumed within 12 h. As shown in Fig. 2c, the ethanol content after the 12 h fermentation of raw DG and TEAC/OA-DG were 2.59 g/L and 10.93 g/L. After enzymatic saccharification and fermentation of TEAC/OA-BDO pretreated DG, the 12 h ethanol production was 16.17 g/L, corresponding to an ethanol yield of 70.87 %. Compared with TEAC/OA-DG, TEAC/OA-BDO pretreatment produced about 1.5 times more the amount of ethanol, which was due to the relatively high content of reducing sugars in the hydrolysate of DG treated with TEAC/OA-BDO. The results indicated that the BDO-assisted TEAC/OA pretreatment was very effective as the green pretreatment with low inhibitors in the fermentation media. Also, it is worth noting that the ethanol concentration showed a decreasing trend after the 24 h fermentation, which may be due to the evaporation of the fermentation product to a certain extent. Fu et al. [34] reported a 73.35 % conversion yield of corn stover hydrolysate in the pretreatment of betaine-glycerol and ChCl-OA. In addition, the ternary TEAC/OA-BDO pretreatment was carried out on *D. sinicus*, *N. affinis*, and *D. Hamiltonian*, which are widely planted in southern China, and these pretreated substrates were subjected to enzymatic saccharification and fermentation. As shown in Fig. 2d, all three pretreated bamboos, like *D. giganteus*, exhibited highly efficient enzymatic hydrolysis and ethanol production. The 72 h glucose production of DS, NA, and DH pretreated with TEAC/OA-BDO was 75.34 %, 68.20 %, and 72.41 %, respectively. Meanwhile, the xylose yields were 80.38 %, 71.52 %, and 77.71 %, respectively. In addition, the glucose released by the enzymatic hydrolysis of three pretreated samples can be converted into ethanol at concentrations of 17.95 g/L, 15.67 g/L, and 16.08 g/L, respectively. The above results indicated that the ternary TEAC/OA-BDO pretreatment is highly effective for the biodegradation of bamboo biomass and has the universality for bamboo.

Furthermore, the material input and output results of cellulose ethanol production were analyzed through bamboo mass balance under the TEAC/OA and TEAC/OA-BDO pretreatment. As shown in Fig. 2e, the raw material contains 54.16 g of glucan, 27.48 g of lignin, and 16.25 g of xylan. After TEAC/OA pretreatment, the glucan recovery in the treated sample was 40.31 g, and cellulase was able to hydrolyze them into 23.69 g of glucose. Meanwhile, 9.78 g regenerated lignin, as a co-product, was obtained from precipitation process. Most glucose can be

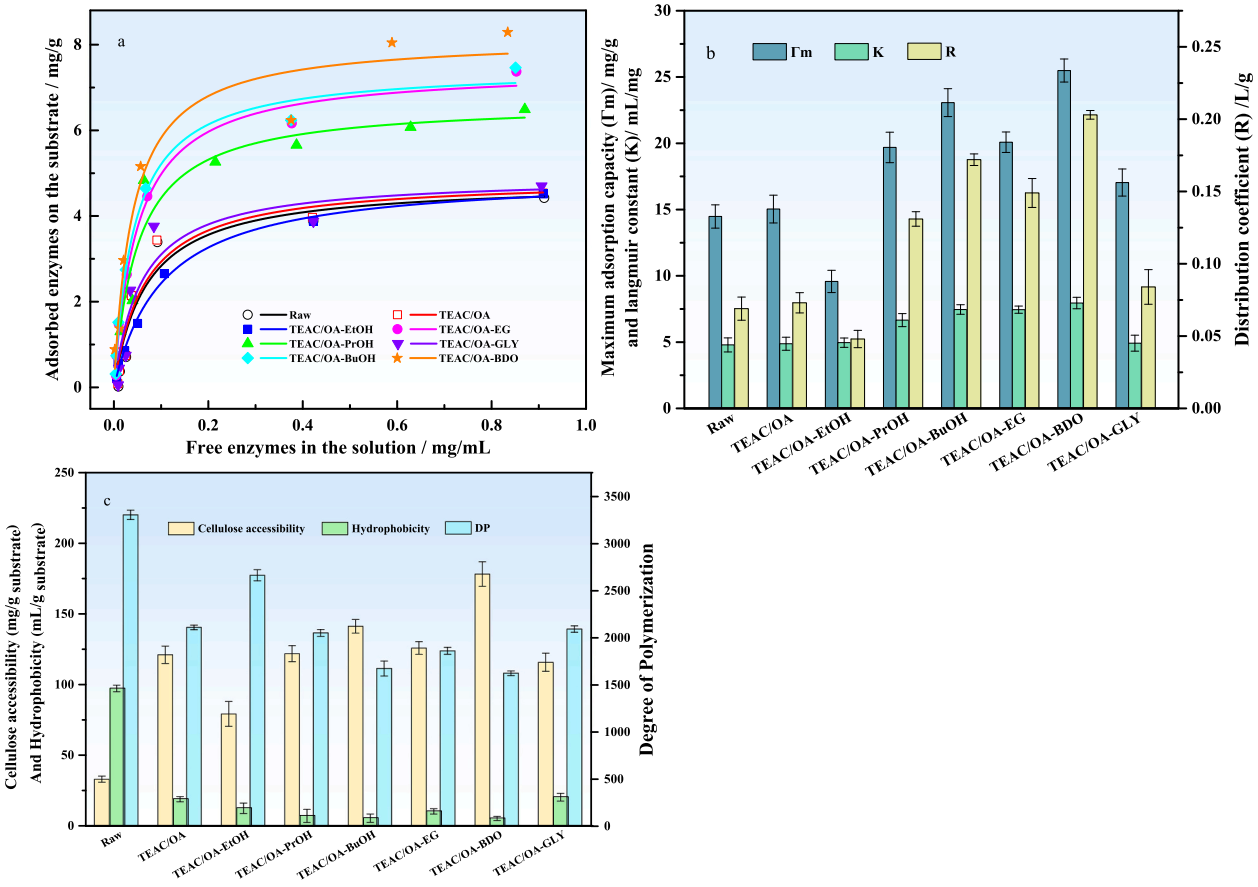


Fig. 3. Enzyme adsorption isotherms (a) and Langmuir adsorption isotherm parameters (b) on untreated and treated *D. giganteus;* and physical properties of pretreated *D. giganteus* (c).

further fermented into 7.78 g ethanol using *Saccharomyces cerevisiae*, suggesting that about 12.85 kg of DG can be turned into 1 kg bioethanol through the binary DES treatment. However, when 100 g of DG was treated with ternary TEAC/OA-BDO, 42.55 g of solid residues were recovered, which contained glucan of 37.39 g, xylan of 2.11 g and lignin of 2.15 g. After the enzymatic hydrolysis and fermentation, bamboo pretreated with TEAC/OA-BDO yielded a total of 28.47 g of fermentable sugars and 10.51 g of ethanol, which was an increase of about 20 % and 32 % over the TEAC/OA treated DG. Besides, 10.43 g of lignin with high β-O-4 content was directly recovered, with a theoretical yield to 37.95 %. Overall, approximately 9.51 kg of DG can be converted into 1 kg of ethanol and 1.01 kg of lignin through TEAC/OA-BDO pretreatment. The obtained sugar-based compounds and ethanol can be further conversed into value-added fuels and chemicals, which is a potential biorefinery process of bamboo biomass.

3.3. Effect of ternary DES pretreatment on enzymatic adsorption and cellulose accessibility of D. giganteus

As shown in Fig. 3a and b, it was obvious that the addition of different polyols greatly improved the adsorption capacity (Γ_m) of treated biomass and their Langmuir constant (K). The value of Γ_m in TEAC/OA pretreated DG were 15.04 mg/g. The addition of polyol to TEAC/OA enhanced the Γ_m value of treated samples to 19.69 mg/g for TEAC/OA-PrOH, and 23.07 mg/g for TEAC/OA-BuOH. The addition of EtOH had a negative impact on enzymatic adsorption. Moreover, the Γ_m value enhanced with the increase of hydroxyl groups in the polyol structure. The TEAC/OA-BDO substrate has the highest maximum adsorption capacity for enzymes (25.49 mg/g), followed by TEAC/OA-EG (20.08 mg/g) and TEAC/OA-GLY (17.04 mg/g), which shown about 10 % higher adsorption capacity (Γ_m) than their corresponding monobasic alcohol. In addition, the K and R values of TEAC/OA-EG and TEAC/OA-BDO (K = 7.45 and 7.79 mL/mg, R = 0.149 and 0.203 L/g) were 1.6–2.1 times higher than those of DG treated with binary DES (4.88 mL/mg and 0.073 L/g). The lower enzyme adsorption capacity of TEAC/OA pretreated substrates may be due to the competitive adsorption of condensed lignin on their surface. The improvement of enzyme adsorption capacity is mainly due to the effective dissociation of barrier components, which provides more production sites for cellulase. The protective effect of polyols on lignin can reduce the deposition of pseudo-lignin and condensed lignin on the surface of cellulose-rich substrates, and further enhanced enzymatic saccharification to release high concentrations of glucose [35]. It meant that these pretreated samples had higher enzymatic hydrolysis efficiency than DG pretreated with TEAC/OA-EtOH, TEAC/OA-PrOH and TEAC/OA-BuOH, which is consistent with the previous enzymatic hydrolysis results.

As expected, the addition of polyols to binary TEAC/OA system also enhanced significantly the accessibility of cellulose to enzymes. In Fig. 3c, raw DG with compact structure had low cellulose accessibility to cellulases of 33.37 mg/g sample. Due to the destruction of DG structure during the binary DES pretreatment, the cellulose accessibility of substrates was 121.39 mg/g substrate, with an increase of 3.6 folds. With the addition of polyols with different carbon chain lengths, the cellulose accessibility of substrates showed an increasing trend. The bamboo pretreated with TEAC/OA-BuOH had the higher cellulose accessibility (141.45 mg/g), followed by that of TEAC/OA-PrOH (122.13 mg/g), suggesting that the addition of polyols effectively enhanced the adsorption capacity of substrates on cellulase. The increase of cellulose accessibility is because of the removal of key barrier lignin, and as lignin is removed, the productive adsorption of cellulose-rich substrates on

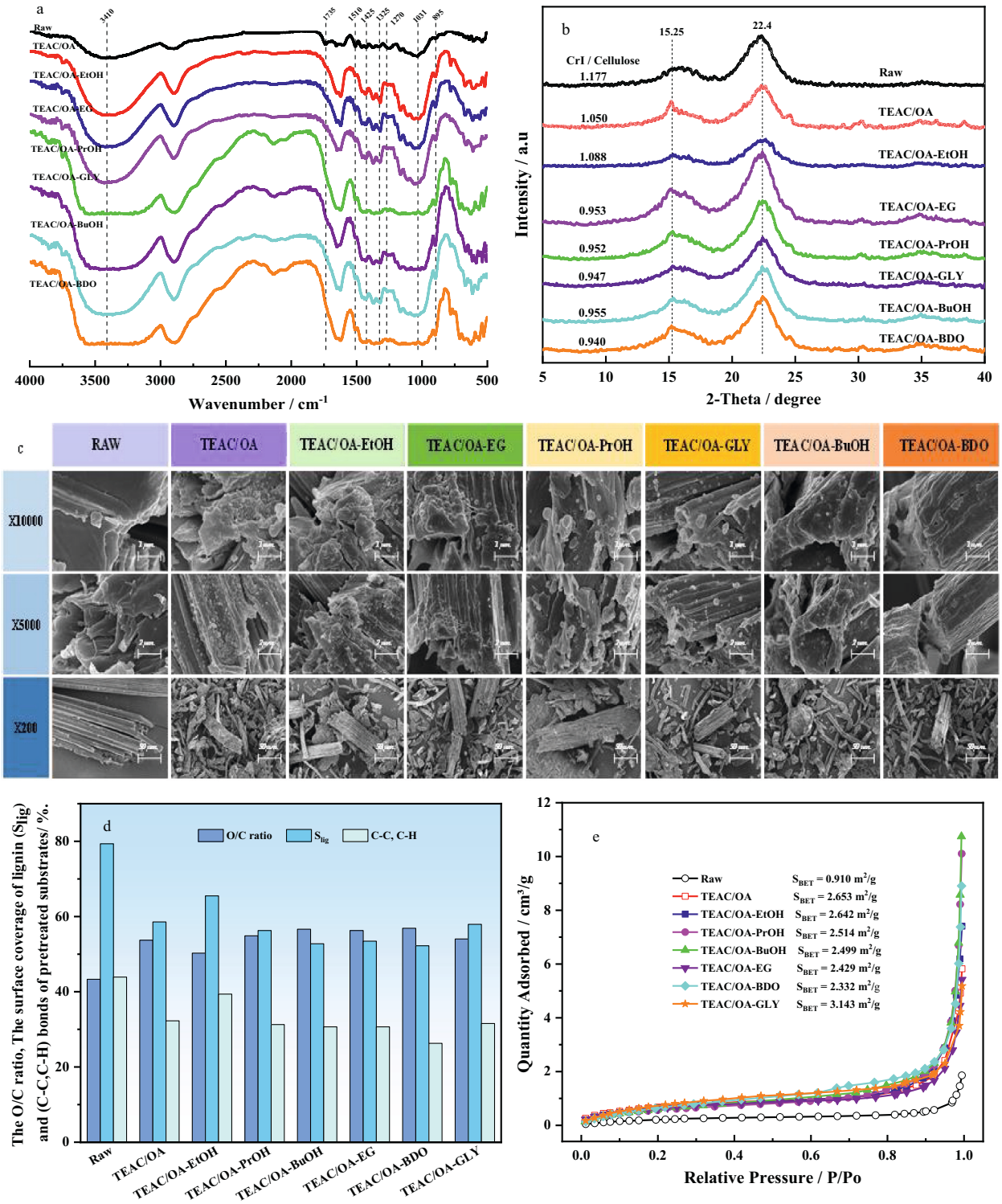


Fig. 4. Structural characteristic of different pretreated substrates. (a) FT-IR spectra; (b) XRD patterns; (c) SEM images; (d) XPS analysis; (e) BET surface area.

cellulase will significantly increase. In addition, the residues after TEAC/OA-BDO, TEAC/OA-BuOH and TEAC/OA-EG pretreatment exhibited a cellulose accessibility, with values of 178.44 mg/g, 141.45 mg/g, and 126.1 mg/g, respectively, which was positively correlated with the hydroxyl content of polyols in DES system. The increase in cellulose accessibility indicated the porosity and looseness of substrate structure, which meant that DES pretreatment caused irreversible damage to the natural recalcitrant structure of bamboo, providing favorable conditions for enzymatic hydrolysis. Usually, lignin and hemicellulose in the lignocellulosic biomass are the two main factors that can inhibit the enzymatic saccharification of cellulose-rich residues, and they crosslink with each other to form a network structure covering cellulose. When there was little significant difference in the content of lignin and xylan, the improvement in enzyme adsorption and hydrolysis efficiency between binary and ternary DES pretreated samples should be because of the presence of coverings on the substrate surfaces, such as pseudo-lignin and condensed lignin. These surface coverings can undergo irreversible non-productive adsorption with cellulase, leading to a

decrease in enzyme activity and thus inhibiting the conversion of carbohydrates into glucose and xylose.

3.4. Effect of ternary DES mixture on the structural characterization of pretreated solid fraction

The changes in bamboo structure and properties after pretreatment were identified as the main factors contributing to the increase in cellulose accessibility and enzymatic digestibility. Raw bamboo and binary DES pretreated DG were also investigated for comparison.

3.4.1. Degree of polymerization and hydrophobicity

Low degree of polymerization (DP) means that more reducing ends in cellulose are exposed after effective glycosidic bond cleavage, which can also improve subsequent enzyme adsorption and saccharification efficiency. In Fig. 3c, the DP of untreated DG was 3307.96, significantly higher than most woody biomass. Upon undergoing the TEAC/OA pretreatment, it dramatically decreased to 2114.73, a 36 % reduction compared with the raw bamboo. When EtOH, EG and PrOH were added to the TEAC/OA system, the DP values of samples further decreased to 2667.37, 2055.68 and 1678.07 respectively. It is observed that TEAC/OA-BDO had a DP value of 1628.07, which was possibly because of the strong swelling effects by DES pretreatment. The hydrophobic properties of raw bamboo and pretreated residues were analyzed in Fig. 3c. Compared with raw bamboo, the hydrophobicity of pretreated bamboo has been reduced to a certain extent, which is consistent with the removal of xylan and lignin. Specifically, the hydrophobicity of trinary DES pretreated solids progressively decreased with increasing the carbon chain length and hydroxyl content in the polyol structure. The hydrophobicity of TEAC/OA-BDO reached 5.60 mL/g, reduced by about 56 % compared to TEAC/OA pretreatment, mainly because of the reduction of the blockage of hydrophobic substances, such as lignin, xylan and their degradation products [36,37]. The increase of hydrophilicity promoted the binding between cellulose substrates and cellulases, ultimately leading to an increase in enzymatic hydrolysis yield.

3.4.2. FTIR spectra

As shown in Fig. 4a, except for certain specific intensity changes, most of the absorption peaks of all samples are similar in the entire wavenumber range, suggesting that both binary and ternary DES pretreatment did not change the chemical structure of bamboo. The peak at 2900 cm^{-1},1104 cm^{-1} and 1029 cm^{-1}, representing the stretching vibration of the methyl C—H, C—O and C—C bonds in cellulose increased, meant that the DES pretreatment can expose more cellulose. The absorption peak at 1730 cm^{-1}, related to carbonyl groups in branched hemicellulose, was weakened or even disappeared after the binary DES pretreatment, suggesting the effective degradation of hemicellulose and destruction of LCC structure during the fractionation, mainly because hemicellulose is highly sensitive to the acid condition, due to its low DP and branched structure. Additionally, the characteristic peaks of lignin at 1510 cm^{-1} and 1425 cm^{-1} in all pretreated samples decreased slightly in intensity. The intensity of peak at 1315 cm^{-1} belonging to S-lignin increased, indicating that the G-lignin is more easily broken during the DES pretreatment, which can be confirmed by the subsequent HSQC NMR analysis. The intensity of the above absorption peaks is strengthened with the increase of hydroxyl groups in the polyol structure, while there is no obvious difference in the intensity of signal peak after the alkyl chain is extended.

The ratio of CrI/cellulose is commonly used to evaluate changes in the crystal structure of pretreated substrates. It can be seen from Fig. 4b that raw bamboo had a CrI/cellulose of 1.177, while the value of TEAC/OA pretreated DG decreased to 1.050 due to the removal of lignin (77.63 %) and xylan (96.21 %). When using ternary DES of TEAC/OA-EtOH, TEAC/OA-PrOH and TEAC/OA-BuOH, the CrI/cellulose value of substrates further decreases to 1.088, 0.952 and 0.955 respectively. Previous studies have shown that the influence of acid-based pretreatment on the biomass crystallinity was mainly because of the removal of amorphous components and the swelling of cellulose crystal structure [38]. The CrI/cellulose ratio of substrates pretreated with TEAC/OA-EG (0.953), TEAC/OA-GLY (0.947), and TEAC/OA-BDO (0.940) was lower than that of the other three ternary DES pretreatments, indicating that the cleavage highly ordered hydrogen bonds of crystalline cellulose and swelling of cellulose components were more intense.

3.4.3. SEM and XPS analysis

The visualized changes in the surface and microstructure of bamboo residues were characterized with SEM techniques. As shown in Fig. 4c, it was evident that the fibers in raw bamboo are tightly wrapped by lignin and hemicellulose, presenting a flat, smooth, and dense structure, which thoroughly hindered the accessibility of cellulase to cellulose. Compared to the raw bamboo, the surface of DES treated samples showed curling and many cracks. After TEAC/OA pretreatment, the fragmentation degree of DG increased, and the surface of cellulose-rich substrates became rougher and looser, indicating a significant depolymerization effect of TEAC/OA on bamboo cell wall. Moreover, some spherical-shaped particles or droplets were observed on the surface of TEAC/OA pretreated DG, which could be caused by the migration of lignin from the internal region to the external surface, as well as the production of pseudo-lignin under severe acidic conditions [39]. Interestingly, compared with binary DES pretreated sample, there were only a small number of lignin droplets in the SEM image of ternary DES, and even no lignin droplets were found on the TEAC/OA-BuOH and TEAC/OA-BDO, suggesting that the addition of polyols contributed to the dissolution of pseudo-lignin and surface lignin on the substrates. These SEM results can also be confirmed by XPS spectroscopy in Fig. 4d. After TEAC/OA pretreatment, the overall O/C ratio improved from 0.43 of raw material to 0.54, while the coverage of surface lignin reduced from 79.34 % to 58.57 %, indicating that the binary DES pretreatment removed most of the lignin and thus improved the O/C ratio. As shown in Fig. 4d, TEAC/OA-PrOH, TEAC/OA-BuOH, TEAC/OA-EG and TEAC/OA-BDO pretreatment overcame the formation and adverse effects of surface lignin, reducing the concentration of surface lignin to 56.27 %, 52.76 %, 53.46 %, and 52.22 %, respectively. Additionally, in the raw and binary DES pretreated DG, the signal intensities from C—C and C—H were 43.91 % and 32.28 %. But the peak intensities of C-C/C-H represented the lignin content in TEAC/OA-PrOH, TEAC/OA-BuOH, TEAC/OA-EG and TEAC/OA-BDO samples were relatively low, at 31.28 %, 30.66 %, 30.66 %, and 26.29 %, respectively, suggesting that the four pretreatment methods can reduce the content of condensed and surface lignin. Compared with TEAC/OA, the content of surface lignin and C-C/C-H in TEAC/OA-BDO were reduced by 11 % and 19 %, respectively, indicating that TEAC/OA-BDO pretreatment has a good ability to inhibit the formation of pseudo-lignin and surface lignin.

3.4.4. BET surface area

To more intuitively demonstrate the changes in the surface morphology of bamboo residues after TEAC/OA pretreatment, BET surface area was also measured to explain the differences in enzymatic hydrolysis efficiency. As shown in Fig. 4e, compared with raw DG (0.910 m^2/g), the BET surface area increased by 2.9 times to 2.653 m^2/g after binary TEAC/OA pretreatment. This is the result of TEAC/OA pretreatment removing most of the hemicellulose and lignin, leading to the opening of the tight structure of bamboo, matrix fragmentation and cavity formation, thereby increasing the surface area of the substrate. However, most ternary DES pretreatments showed a certain degree of reduction in BET surface area compared to TEAC/OA. The surface area of TEAC/OA-EG and TEAC/OA-BDO were 2.429 m^2/g and 2.332 m^2/g respectively, which is approximately 12 % less than that of TEAC/OA. The decrease in surface area is closely related to the reduction in content of surface lignin, which may be because the addition of polyols improved the lignin solubility and hindered the redeposition of lignin on the surface of cellulose [40].

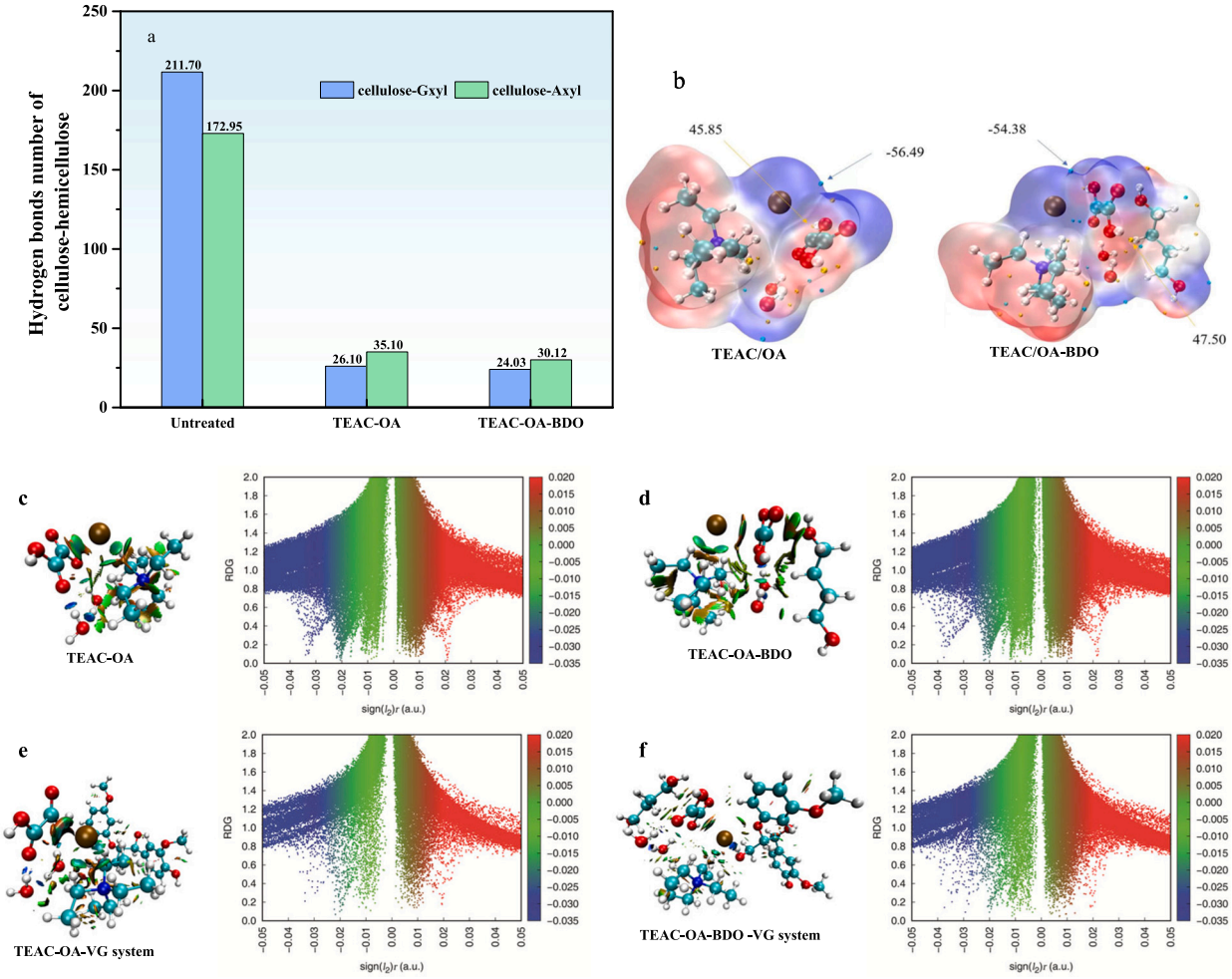


Fig. 5. Calculated intermolecular interactions between DES system and cellulose-hemicellulose and cellulose-lignin model compounds. (a) hydrogen bonds number change of cellulose-hemicellulose in the different DES system; (b) electrostatic potential plot of TEAC/OA and TEAC/OA-BDO; (c-f) scatterplot of noncovalent interactions between DES system and lignin model compounds.

3.5. Theoretical and computational study on the interaction mechanism between bamboo residue and ternary DES

The hydrogen bonds interaction between the DES solvents and cellulose-hemicellulose system was analyzed using classic molecular dynamics simulations. As showed in Fig. 5a, the hydrogen bond number of cellulose-Gxyl and cellulose-Axyl system was 211.7 and 172.95, suggesting that hemicellulose and cellulose are closely connected through intermolecular hydrogen bonding. The addition of binary and ternary DES significantly changed the hydrogen bond network of the original two systems. After adding TEAC-OA DES, the number of hydrogen bonds in the cellulose-Gxyl and cellulose-Axyl decreased to 26.10 and 35.10, respectively. The TEAC/OA-BDO system has higher hydrogen bonding ability with hemicellulose-cellulose system, which reduced the number of original hydrogen bonds by 8.8- and 5.74-times. This is mainly because the hydrogen bonds in DES can interact with hydroxyl groups (− OH) in cellulose and hemicellulose, as well as ethers and hydroxyl groups in lignin. These interactions help break the intramolecular and intermolecular hydrogen bonds in biomass, making cellulose, hemicellulose, and lignin easier to separate [41]. DES with a high number of hydrogen bonds can dissolve lignin and hemicellulose more effectively by forming competitive hydrogen bonds with these components. Therefore, the ternary DES has slightly stronger damage to the hydrogen bonds in cellulose-hemicellulose than the binary DES.

The electrostatic potential surfaces and geometric structures of TEAC/OA and TEAC/OA-BDO are shown in Fig. 5b, where the purple, white and red parts represented negative, zero, and positive electrostatic potential, respectively. The maximum electrostatic potentials of TEAC/OA and TEAC/OA-BDO are −56.49 kcal/mol and -54.38 kcal/mol, respectively, suggesting that the negative potential of the trinary DES solvent was strong and mainly around Cl^-. Meanwhile, the hydrogen proton of 1,4-butanediol in TEAC/OA-BDO has a strong positive potential (47.50 kcal/mol), which formed strong hydrogen bonding interactions with Cl^- in HBA (TEAC), leading to a greater degree of delocalization of the hydrogen proton. This is very important for breaking the hydrogen bonding network structure of lignocellulosic biomass. Figs. 5c-f depicted the RDG analysis and scatter plots of two DES solvents and DES-VG systems. The hydrogen bonds are characterized by blue, while van der Waals interactions and steric repulsion are displayed in the red and green regions, respectively [22]. As shown in Fig. 5c and d, van der Waals forces and electrostatic interactions coexisted in TEAC/OA and TEAC/OA-BDO systems. And the ternary DES system exhibits significant van der Waals forces. After comparing with Fig. 5c-f, it was found that the addition of DES significantly reduced the intermolecular interaction energy of the cellulose-lignin system, indicating that DES can effectively promote the separation between cellulose and lignin. Moreover, ternary DES seems to be more conducive to the interaction with the cellulose-lignin system. These results indicated that the addition of BDO changed the number and strength of hydrogen bonds, as well as the electrostatic potential in the TEAC/OA system, thereby promoting the separation of lignocellulosic biomass and providing some theoretical supports for the experimental results.

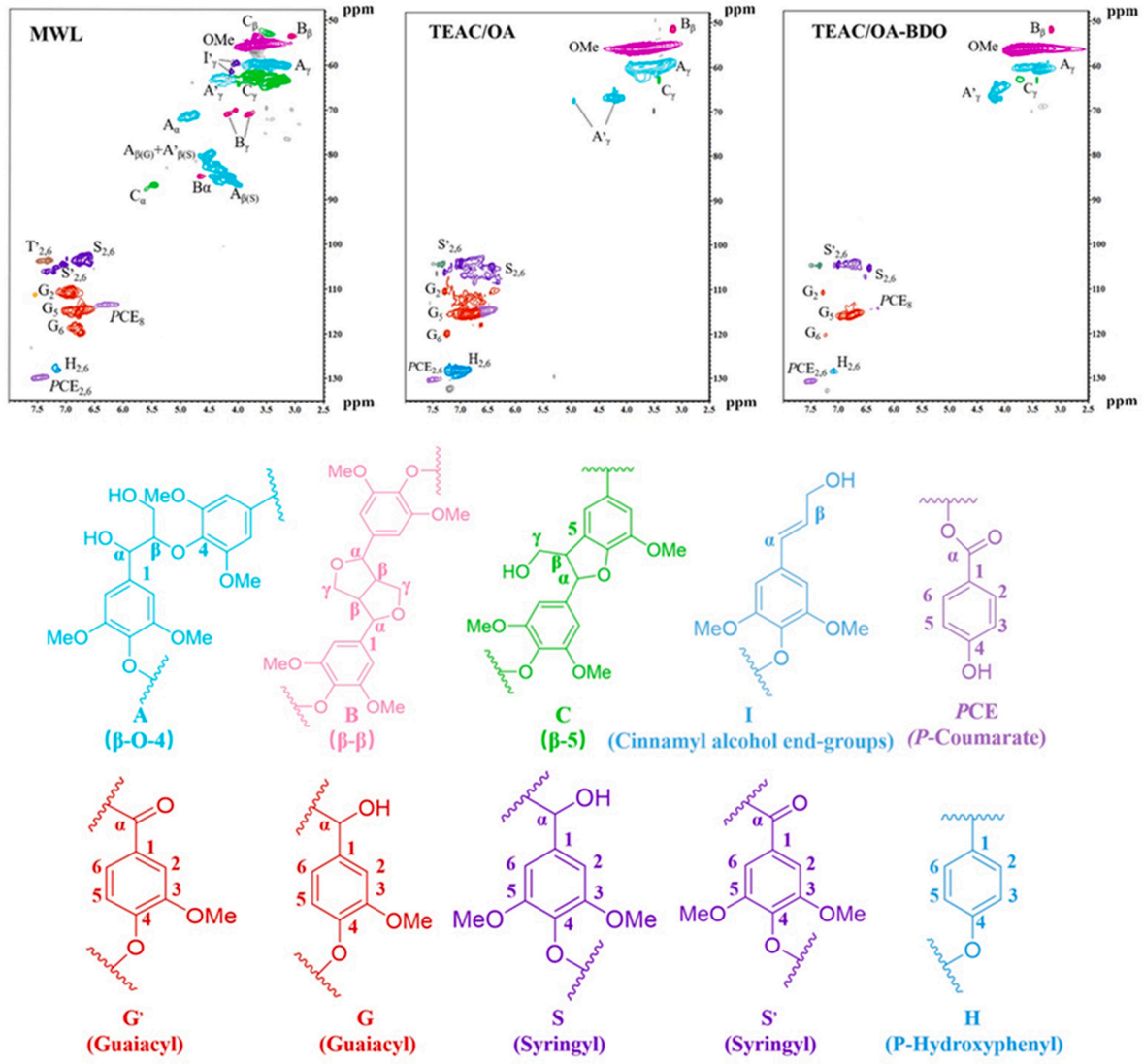


Fig. 6. The 2D-HSQC spectra and main structures of MWL and recovered lignin from TEAC/OA and TEAC/OA-BDO.

Table 1
The different OH groups in DES regenerated-lignin.

Sample	Aliphatic OH (mmol/g)	Syringyl OH (NC, mmol/g)	Guaiacyl OH (mmol/g)		P-hydroxyl OH (mmol/g)	Carboxylic group (mmol/g)	Total phenolic OH (mmol/g)	S/G ratio (phenolic OH)
			C	NC				
MWL	4.67	0.06	0.09	0.43	1.21	0.12	1.91	0.12
TEAC/OA	0.32	0.52	0.11	0.32	0.82	0.51	2.28	1.21
TEAC/OA-BDO	0.63	0.58	0.15	0.36	1.18	0.37	2.64	1.14

3.6. Characteristics of regenerated-lignin from DES pretreatment

During the DES pretreatment of bamboo, a large amount of lignin in the raw material can be extracted into the pretreatment solution. In order to make the whole bamboo biorefinery process economically feasible, the lignin in the liquid fractions should also be effectively recovered and fully utilized. The delignification and the lignin recovery yields by TEAC/OA pretreatment were 77.63 % and 40.33 %, respectively, while those by TEAC/OA-BDO were 91.98 % and 42.60 %, respectively (Figs. 1 and 2). It was obvious that the yields of the two DES-lignin were lower than the corresponding delignification, mainly because the acid soluble lignin dissolved in DES were not easily regenerated. And the chemical structural characteristics of regenerated-lignin were comprehensively studied through 2D HSQC (Fig. 6) and ^{31}P quantitative NMR analysis (Table 1). The 2D NMR spectra are divided into side-chain (δ_C/δ_H = 95–45 ppm/6.5–2.0 ppm) and aromatic (δ_C/δ_H = 160–90 ppm/8.0–6.0 ppm) regions, with the main lignin cross-signals labeled according to previous literature [42,43]. As illustrated in the side chain region, MWL exhibits obvious signals of β-β (B_α), β-5 (C_α), β-O-4 (A_α) and methoxy groups (OMe) linkages at 83.63/4.55, 88.03/5.52, 70.6/5.05 and 56.29/3.75 ppm. And γ-acetylated β-O-4 bond ($A_{\gamma'}$) was observed, suggesting the presence of LCC compounds in DG. After TEAC/OA and TEAC/OA-BDO pretreatment, it can be noticed that the signals of β-O-4 bond, A_α and A_β are weakened or even disappeared

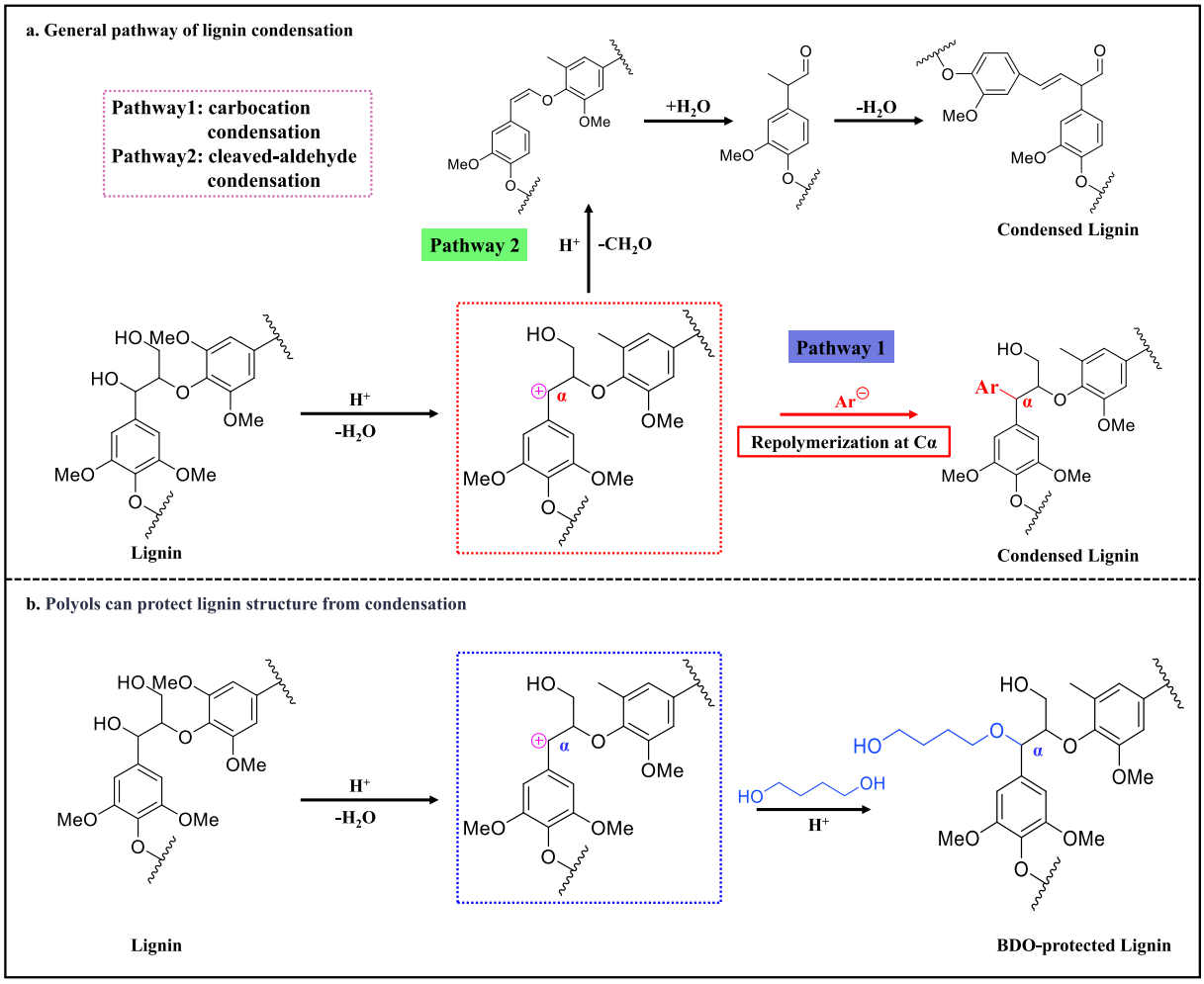


Fig. 7. Reaction pathways of lignin in binary and ternary DES pretreatment processes.

compared to MWL. The degradation of lignin macromolecules was mainly due to the catalytic cleavage of β-O-4 bonds by hydrogen proton in the strong acidity of TEAC/OA. It was found that β-β bonds in resinol (B) were present in both TEAC/OA- and TEAC/OA-BDO-lignin, suggesting the cleavage of β-O-4 aryl ether linkages and condensation could occur simultaneously. In addition, a new peak was observed at δ_C/δ_H 65.24/4.29, indicating that esterification reaction occurred at the C_α position of lignin, forming a new side chain structure [44]. In the aromatic region of regenerated-lignin, the main signals, such as guaiacyl (G), syringyl (S), and 4-hydroxyphenyl (H) units, were obviously recognized in bamboo MWL. After the TEAC/OA pretreatment, the condensed signals of $S_{2,6}$ and G_5 lignin were detected in the HSQC spectrum, mainly due to the attack of α-carbocations formed by lignin on the nucleophilic sites of lignin. However, these lignin condensation signals were weakened during the BDO-assisted TEAC/OA pretreatment. Previous studies have shown that incorporation of polyols into lignin can inhibit the recondensation reaction during the lignin fractionation [45]. Additionally, the attachment of BDO to lignin side chains also increased the steric hindrance of molecules, thus inhibiting the formation of condensation structures. The TEAC/OA-BDO system achieved the satisfactory yield of lignin removal and low condensation reaction, which can greatly promote the large-scale production of this high-quality.

The changes of various hydroxyl groups in the lignin structure were investigated using the ^{31}P NMR (Table 1). The content of aliphatic OH and total phenolic OH in MWL is 4.67 mmol/g and 1.91 mmol/g, respectively. After pretreatment with binary and ternary DES, the content of aliphatic OH significantly decreased to 0.32 mmol/g and 0.63 mmol/g, while the total phenolic OH increased to 2.28 mmol/g and 2.64 mmol/g, respectively, indicating rapid dehydration and condensation occurred during the β-O-4 bond cleavage process. This may also be due to the incorporation of diol molecules into the α site of lignin in the TEAC/OA-BDO pretreatment. Table 1 also shows a significant increase in the content of non-condensed S-OH groups, indicating that the β-O-4 ether linkage was very easily dissociated in the DES system. We also observed that the non-condensed OH content of TEAC/OA-BDO lignin was significantly higher than that of TEAC/OA lignin, which may provide evidence for BDO to prevent lignin condensation during the pretreatment. The increase of more active OH can enhance the applicability of lignin-based materials and polymers synthesized from lignin. Besides, the S/G ratio of lignin can also reflect the structural changes during the pretreatment process. In Table 1, the S/G ratio of MWL is 0.12, lower than the values of TEAC/OA lignin (1.21) and TEAC/OA-BDO lignin (1.14). The increase in S/G ratio is because DES pretreatment facilitated the preferential dissolution of S units from the cell wall, while G lignin units are more prone to condensation.

In the work, 1,4-butanediol (BDO) was found to be more effective than other polyols in protecting lignin structures from condensation during the DES pretreatment. Based on the experimental results and the previous literature, a possible protective mechanism of polyols on lignin structure could be proposed. As shown in Fig. 7a, the LCC structure of biomass and the ether bond linkages of lignin were attacked by hydrogen protons during the acidic DES pretreatment, leading to the breakage of α-ether bonds and the formation of α-carbocation. These

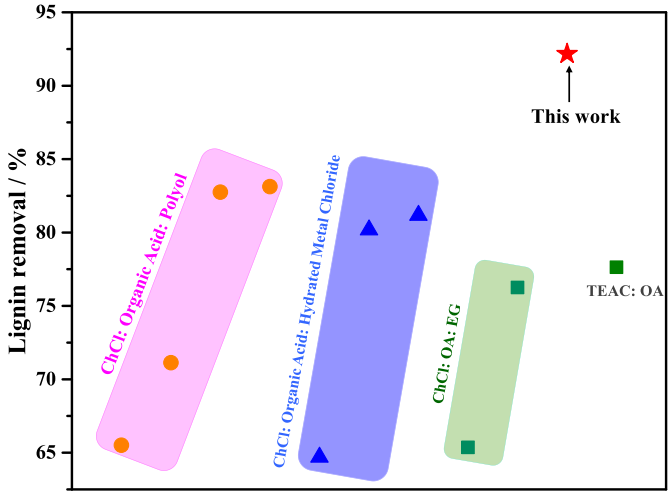


Fig. 8. Comparison of delignification obtained in this study and the previous studies.

active C_α can undergo intramolecular condensation reactions with some adjacent aromatic groups of lignin, forming condensed lignin (Pathway 1) [46,47]. Another way that the intermediate of α-carbocation can also be further dehydrated to form aldehydes, which then undergo condensation reactions with alcohol hydroxyl groups on lignin, further condensing the lignin structure (Pathway 2). At the same time, a series of side condensation reactions inevitably occurred at high temperature. However, during the trinary DES pretreatment, polyols as a nucleophile reagent can attack the carbocation intermediate and form α-etherified lignin, thereby preventing and inhibiting lignin condensation (Fig. 7b) [48,49]. Previous reports have shown that diol compounds can assist acidic DES mixtures in the biomass separation. Ee et al. designed a DES composed of alanine, lactic acid, and ethylene glycol to separate high-purity lignin, and found that the addition of ethylene glycol can stably avoid lignin condensation [45]. It had reported that the diols-based DES can achieve short-term (10 min) bamboo fractionation [49]. The higher delignification efficiency of 83.12 % was achieved by Choline chloride (ChCl)/lactic acid (LA) with γ-valerolactone (GVL) pretreatment at 130 °C for 4 h [50]. Ethanol assisted choline chloride/lactate DES pretreatment can achieve synergistic carbohydrate conversion and efficient delignification. Ethanol assisted addition facilitates the incorporation of ethyl and reduces the condensation structure of cyclohexenone in lignin structure [51]. Previous studies about various ChCl-based DES pretreatments on lignin removal were shown in Fig. 8. The delignification of TEAC/OA-BDO pretreatment is at the highest level among all DES pretreatment methods [50,52–57]. Compared with choline chloride, TEAC is an inexpensive and easily available HBA that can be used for synthesizing DES solvents for lignocellulose pretreatment. This TEAC/OA-polyol DES pretreatment might provide a solution for the bamboo pretreatment and some guidance for lignocellulosic biorefinery.

4. Conclusions

This study proposes a polyol-assisted ternary DES pretreatment for fractionating bamboo residue, in which the TEAC/OA-BDO system achieved the xylan removal of 86.7 % and delignification of 91.98 %. The subsequent enzymatic digestibility process exhibited a high glucose yield of 76.13 %, which was further converted into 16.17 g/L ethanol using *Saccharomyces cerevisiae*, increased by 47.9 % than the binary TEAC/OA pretreatment. The extracted lignin had the features of higher activity OH and fewer condensation structures. The results revealed that the TEAC/OA-BDO can not only break the ether bond linkages of lignin and LCC complexes, but also avoid the condensation between lignin molecules. In addition, it was found that the delignification and enzymatic saccharification efficiency of bamboo residue are closely related to the lengths of alkyl carbon chains and hydroxyl number in polyols, which can provide important references for the selection of HBD in acidified DES.

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CRediT authorship contribution statement

Raoping Zhao: Writing – review & editing, Writing – original draft, Methodology, Investigation. **Yu Sheng:** Writing – review & editing, Writing – original draft, Resources, Methodology. **Haiyan Yang:** Visualization, Conceptualization. **Zhengjun Shi:** Validation, Formal analysis. **Dawei Wang:** Data curation, Conceptualization. **Jing Yang:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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